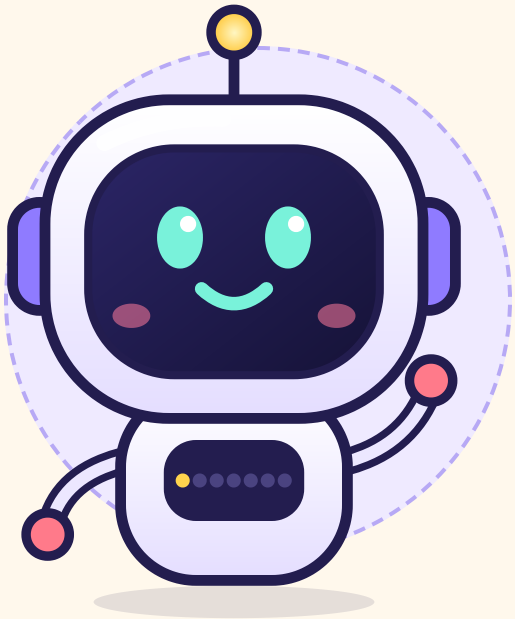
Section 1 · Lesson 1 of 7

Meet Pip: What is AI?

A ready-to-teach, print-and-go kit for ages 6–10. Kids meet Pip, a tiny AI, learn that AI is a computer program that learns from examples, then hunt for AI around a house.



> **"Hi! I'm Pip. I'm a tiny AI. That means I'm a computer program that can learn."**
> From Pip's story, Lesson 1

30–40 min	Unplugged	2 worksheets	Answer key
lesson plan	"AI or Not?" card sort	K–2 and 3–5	+ parent letter

Inside: Teacher guide · Unplugged activity & cards · Worksheet K–2 · Worksheet 3–5 · Answer key · Parent letter · What's next

Created by Nick Castro · howaiworksforkids.com

TEACHER GUIDE

Meet Pip: What is AI?



Learning objective

Students will be able to say, in their own words, that **AI is a computer program that learns patterns from lots of examples, then makes guesses**, and sort everyday things into "uses AI" and "no AI."

Timing plan (about 35 min)

5 min	**Warm-up.** Ask: "What do you think AI is?" Collect ideas on the board, no right or wrong yet. Keep the list to revisit at the end.
5 min	**Read the story on the site.** Project Lesson 1 at howaiworksforkids.com/lessons and read Pip's 5-page picture story together, or tap **"Read it to me."**
7 min	**Play "AI Detective."** Search the room on screen. For each of the 8 things, the class votes: *"Yes, it learns!"* or *"No AI."* Pre-readers can tap **"Hear how to play."** Read Pip's reason aloud after each tap.
12 min	**Unplugged: "AI or Not?"** Card sort (page 3). Groups sort, then explain one card using "It learns from examples..." or "It does the same thing every time..."
5 min	**Wrap-up.** Answer the lesson's 2 quick questions together (on the site, or orally; see answer key), earn the **AI Spotter** badge, revisit the warm-up list.

Stretch to 40 min: add a worksheet (K–2 or 3–5). Short on time: skip the card sort and send the worksheet home with the parent letter.

Materials

- Projector or screen with a web browser (no login needed)
- "AI or Not?" cards: 1 set per group of 3–4, cut out
- Worksheets (page 4 or 5), pencils, crayons; scissors & glue (optional)

Key words

AI (ay-eye) A computer program that learns patterns from examples and then makes guesses.

Example One thing you show an AI so it can learn, like one picture of a cat.

Pattern Something that shows up again and again. Cats have pointy ears. That's a pattern.

Background for you

AI (artificial intelligence) here means a computer program that learns from examples instead of following only rules a person wrote. Shown lots of cat pictures, it finds what is the same (pointy ears, whiskers, a long tail). That repeated thing is a **pattern**. Shown a new picture, it makes a **guess** (grown-ups say "prediction"). Guesses are often right, but not always. A lamp or a wind-up clock does the same thing every time; it never learns or guesses, so it isn't AI, even if it uses electricity.

Discussion prompts

- How did Pip learn what a cat looks like?
- Pip said, "I'm right a lot. But not always." Why might Pip get it wrong?
- What makes a smart speaker different from a lamp?
- What is one thing at home you think might use AI? How could we find out?

Common misconceptions

"AI is alive."	It's a computer program. It can sound friendly, but it has no feelings. Pip: "I'm not alive."
"AI is magic."	It learns from lots of examples people give it. Pip: "I'm not magic."
"AI knows everything."	It only makes guesses from patterns it has seen, and is "right a lot, but not always."
"Anything with a plug or screen is AI."	A lamp uses electricity but never learns. AI is the stuff that learns and guesses.

Concept alignment

AI4K12 Five Big Ideas in AI: Big Idea 3, *Learning* (computers can learn from data), with a touch of Big Idea 1, *Perception* (face unlock, smart speakers). **CSTA K–12 CS Standards:** the *Impacts of Computing* concept (computing in everyday life). Concept-level only; no specific standard codes are claimed.

UNPLUGGED ACTIVITY · WHOLE CLASS OR GROUPS · 12 MIN

AI or Not? Card Sort



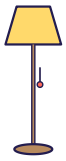
1. Cut out the 8 cards and the 2 sorting labels. These are the same 8 things kids find in the AI Detective game.
2. Place the labels on a table or the board. Hold up each card and ask: “Does it learn from examples and make guesses?”
3. Students sort the card and give a reason: “It learns…” or “It does the same thing every time.”
4. Check with the answer key (page 6). Bonus: draw something from home on a blank card and sort it together.

Yes, it learns!

Uses AI

No AI

Same thing every time



Lamp

Does it learn and guess?



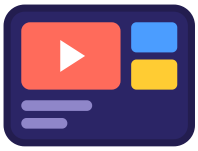
Smart speaker

Does it learn and guess?



Teddy bear

Does it learn and guess?



Video app

Does it learn and guess?



Wind-up clock

Does it learn and guess?



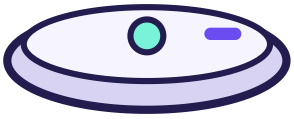
Face-unlock phone

Does it learn and guess?



Books

Does it learn and guess?



Robot vacuum

Does it learn and guess?

Draw your own

Draw your own

Draw your own

Draw your own

✂ Cut along the dashed lines.

Name _____ Date _____

WORKSHEET · GRADES K–2

AI Detective

1 **Circle** the things that **learn and guess**. They use AI! (Hint: there are 4.)


Lamp


Smart speaker


Teddy bear


Video app


Wind-up clock


Face-unlock phone


Books


Robot vacuum

2 **How does AI learn?** Circle one.


From lots of examples


It reads your mind


It's magic

3 **Does AI have feelings like you do?** Circle one.

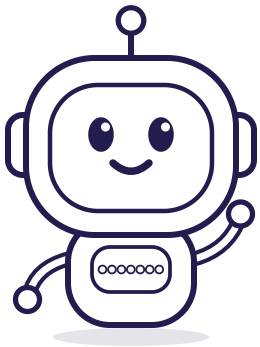

Yes, it gets sad


No, it's a computer program



Only on Tuesdays

4 **Color Pip!** Pip is a tiny AI.

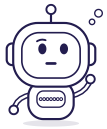


5 **Draw** one thing at home that might use AI.

Name _____ Date _____

WORKSHEET · GRADES 3–5

AI Detective: Sort & Explain



Part A · Sort it. Write each thing from the word bank in the right column.

Lamp · Smart speaker · Teddy bear · Video app · Wind-up clock · Face-unlock phone · Books · Robot vacuum

🧠 Uses AI learns from examples & guesses	🚫 No AI does the same thing every time

Part B · Fill in the blanks. Use: **examples** **pattern** **guess**

1. People show Pip lots and lots of _____, like a thousand pictures of cats.
2. Pointy ears, whiskers, a long tail: what's the same is called a _____.
3. When Pip sees a new picture, Pip makes a _____. It's right a lot, but not always.

Part C · Quick check. Fill in the circle.

4. How does AI learn?

○ From lots of examples ○ It reads your mind ○ It's magic

5. Does AI have feelings like you do?

○ Yes, it gets sad ○ No, it's a computer program ○ Only on Tuesdays

Part D · Explain.

6. Pick one card you sorted as **Uses AI**. How does it learn or guess?

7. Pip says, "I'm not magic. I'm not alive." What is Pip, really?

8. Name one thing at home that might use AI. How could we find out?

FOR TEACHERS

Answer Key

"AI or Not?" cards · AI Detective (the 8 game items)

Reasons are Pip's own words from the AI Detective game. When the game ends, Pip says: "Case closed! You found all 4 AI things. AI is the stuff that learns and guesses."

	**Smart speaker**	✔ **Uses AI**	It learned what words sound like, so it can guess what you asked for.
	**Video app**	✔ **Uses AI**	It looks at what you watched and guesses what you'll like next.
	**Face-unlock phone**	✔ **Uses AI**	It learned what your face looks like from examples, so it can tell it's you.
	**Robot vacuum**	✔ **Uses AI**	It learns a map of the room so it can go around the couch.
	**Lamp**	✘ **No AI**	No AI. You flip the switch and it turns on. It never learns or guesses.
	**Teddy bear**	✘ **No AI**	No AI. A teddy bear doesn't learn anything. It's great for hugs, though!
	**Books**	✘ **No AI**	No AI. A book says the same words every time. People wrote them.
	**Wind-up clock**	✘ **No AI**	No AI. Gears turn the same way every second. No examples, no guessing.

Worksheet K–2

1. Circle: Smart speaker, Video app, Face-unlock phone, Robot vacuum. Not circled: Lamp, Teddy bear, Books, Wind-up clock.

2. 📚 From lots of examples

3. 💻 No, it's a computer program

4–5. Any coloring and drawing. Invite kids to say *why* their thing might use AI.

Worksheet 3–5

A. Uses AI: Smart speaker, Video app, Face-unlock phone, Robot vacuum. No AI: Lamp, Teddy bear, Books, Wind-up clock.

B. 1 examples · 2 pattern · 3 guess

C. 4 From lots of examples · 5 No, it's a computer program

D. Answers vary. Look for: **6** learns from examples/makes a guess (see reasons above); **7** a computer program that finds patterns and makes guesses ("a really good pattern finder"); **8** any reasonable idea + a way to check (ask a grown-up, read about it, see if it learns or guesses).

The 2 quick-check questions (as on the site)

1. How does AI learn?

✔ 📚 **From lots of examples.** Pip's reply: "Yes! AI learns by looking at lots of examples."

2. Does AI have feelings like you do?

✔ 💻 **No, it's a computer program.** Pip's reply: "Right! AI can sound friendly, but it doesn't have feelings."

TAKE-HOME LETTER

Today we met Pip, a tiny AI!



Dear Families,

Today our class started learning how artificial intelligence (AI) works with **Lesson 1, "Meet Pip: What is AI?"** from the free site **howaiworksforkids.com**.

What we learned

"AI is a computer program that learns patterns from lots of examples. Then it makes guesses."

- Pip learns from lots of examples, like a thousand pictures of cats, and finds a **pattern** (pointy ears, whiskers, a long tail).
- Then Pip makes a **guess**. It's right a lot, but not always.
- AI is not magic and not alive. It's a computer program, a really good pattern finder.
- We played **AI Detective**: a smart speaker, a video app, a face-unlock phone, and a robot vacuum use AI. A lamp, a teddy bear, books, and a wind-up clock don't.

Why the site is safe for kids

- **Free**, with **no ads** and no in-app purchases.
- **No accounts.** Lesson progress is saved only in your browser and never sent anywhere.
- **No kid data.** It never asks for a child's name, email, school, photo, or location.
- **No chatbot.** Kids can't type to an AI. Every answer in the games was written in advance.
- The site counts anonymous visits with Google Analytics, with ad features turned off. Details: howaiworksforkids.com/privacy

3 questions to talk about at home

1. **What is one thing at home you think might use AI? How could we find out?**
2. **How did Pip learn what a cat looks like?**
3. **Pip is "right a lot, but not always." Can you think of a time a phone, speaker, or app guessed wrong?**

Want to try it together? Visit **howaiworksforkids.com**. Each lesson takes about 5 minutes, works on phones, tablets, and computers, and needs no sign-up.

Thank you!

_____ (Teacher)

COMING SOON

Want all of Section 1?

The full **How AI Works for Kids Teacher Pack** will bring this same print-and-go format to every lesson: a teacher guide, an unplugged activity, K–2 and 3–5 worksheets, an answer key, and a parent letter.



1. **Meet Pip: What is AI?** — Game: AI Detective · Badge: AI Spotter — This kit
2. **Teach by Showing** — Game: Sort the Glorbs · Badge: Pip's Teacher
3. **How Chatbots Talk** — Game: Next Word! · Badge: Word Wizard
4. **Good Examples In, Good Guesses Out** — Game: Fix Pip's Mix-up · Badge: Clue Fixer
5. **Sure Doesn't Mean Right** — Game: Fact or Fib? · Badge: Fib Finder
6. **Real or Made by AI?** — Game: Spot the Glitches · Badge: Glitch Spotter
7. **Be the Boss of AI** — Game: Go, Ask, or Stop · Badge: AI Boss

Watch for the Teacher Pack at:

howaiworksforkids.com/teachers

All 15 lessons on the site are free for classrooms and stay free. No ads, no accounts, no chatbot, no kid data.

Created by Nick Castro · howaiworksforkids.com